

Full Length Article

CoManTrack: Conflict-aware and manifold-adaptive RGBT tracking

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ABSTRACT

RGBT tracking relies on fusing visible and thermal signals, but severe degradation in one modality often pollutes the shared feature space, causing cascading errors and long-term temporal drift. Existing methods struggle with active thermal distractors and suffer from over-smoothed representations and poisoned memory queues during online adaptation. To address these issues, we propose CoManTrack, a spatio-temporally decoupled framework that isolates spatial purification from temporal adaptation. Spatially, we introduce Conflict-Driven Directed Dampening (CD3) to dynamically suppress high-energy thermal noise based on semantic divergence, safeguarding fragile visible structures. The purified features are then enriched by a Tri-Statistic Feature Envelope (TSFE), which captures fine-grained multi-granularity cues through normalized extreme statistics. Temporally, we develop Test-Time Optimal Manifold Adaptation (TOMA) to filter redundant historical states based on distributional novelty. This maintains a diverse and compact target manifold, ensuring a stable geometric foundation for dynamic adaptation. Extensive experiments on the LasHeR, RGBT234, and GTOT benchmarks demonstrate that our CoManTrack achieves competitive and balanced performance against state-of-the-art methods, validating its robustness against severe cross-modal degradation.

1. Introduction

Fusing visible and thermal signals is a standard practice to maintain tracking stability under harsh conditions like darkness or heavy clutter (Lu et al., 2025b). While this bimodal integration theoretically guarantees complementary perception, complex real-world dynamics frequently break this assumption (Wang et al., 2024b). If one modality becomes severely degraded, standard cross-modal interactions often fail to isolate the noise, inadvertently allowing local spatial distortions to pollute the shared feature space (Yue et al., 2024). Ultimately, these spatial errors cascade into the temporal dimension. As corrupted frames continuously update the tracker's memory, the accumulated noise inevitably compounds over time, forcing the tracking trajectory to drift (Chen et al., 2024).

Recent trackers attempt to resolve these cascading errors through cross-modal attention and temporal memory queues (He et al., 2024; Tang et al., 2024a). However, these generic countermeasures often fall short in extreme scenarios because they ignore the distinct physical properties of each modality. Standard symmetric fusion treats visible and infrared streams as equals, failing to recognize that aggressive thermal distractors can actively hijack the shared semantic space, whereas visible degradation is mostly passive (Zhang et al., 2024c).

This spatial corruption is further exacerbated by conventional feature aggregation, which over-smooths the representations and discards the fine-grained, multi-granularity details essential for robust localization. Consequently, traditional Test-Time Adaptation (TTA) frameworks naively absorb these compromised spatial features (Shao et al., 2025b). Blind to distributional novelty, chronological memory queues become rapidly poisoned by temporal redundancy when the target stalls. This redundancy not only triggers catastrophic forgetting during rapid deformations but also generates ill-conditioned matrices that fundamentally cripple downstream parameter recovery.

Driven by these critical insights, we propose CoManTrack, a spatio-temporally decoupled RGBT tracking framework that treats modality degradation as a cascading process from spatial contamination to temporal drift. Spatially, to realize the conflict-aware design, we treat thermal distractors as active threats to the bimodal feature space. By deploying a directed dampening mechanism, we explicitly suppress high-energy infrared noise and protect fragile visible structures. The purified spatial representations are subsequently enriched through a multi-granularity perception paradigm; moving beyond singular mean-pooling prevents cross-modal alignment from acting as a low-pass filter, thereby preserving the target's complete statistical envelope. Temporally, to achieve manifold-adaptive tracking, we replace passive memory queues with

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a dynamic, distribution-aware evaluation mechanism. By assessing the statistical novelty of incoming observations to filter out redundant states, we maintain a diverse and well-conditioned target representation, effectively preventing memory collapse and driving robust online adaptation. In this way, spatial purification, statistical enrichment, and temporal adaptation form a functionally coupled processing chain rather than a set of independently stacked modules.

To materialize the spatial-temporal decoupling philosophy, we first address spatial modality conflicts through a directed dampening strategy. Specifically, a global conflict score is computed based on the macroscopic semantic divergence between the visible and infrared streams. This metric directly drives a dynamic, temperature-scaled gating function. Through this structural dependency, the network actively suppresses high-energy infrared noise while preserving the integrity of the foundational visible features. Following this purification, the spatial representations are enriched by extracting multi-granularity statistics. By synchronously computing the minimum, mean, and maximum values at the token level, the framework captures highly discriminative peak responses alongside global contextual cues. To manage the variance explosion inherent in aggregating these extreme statistics, a pre-scale normalization mechanism standardizes the overall feature distribution, ensuring stable cross-modal alignment without sacrificing fine-grained structural details.

For the temporal processing phase, we tackle the recency bias typically found in standard memory queues by introducing a dynamic on-line adaptation mechanism. Instead of simply storing frames chronologically, we capture historical target appearances as individual Gaussian distributions using their batch normalization statistics. By calculating the 2-Wasserstein distance, we can accurately measure how novel an incoming frame is compared to the stored history. If a new observation is redundant, we compress it into the existing memory using barycenter refinement; if it provides distinct information, it is selectively added. Alongside this, we permanently secure the initial ground-truth template to guard against long-term tracking drift. Ultimately, this selective memory update builds a highly diverse and well-conditioned difference matrix. This structured diversity ensures that applying singular value decomposition later will reliably extract the true directions of domain shifts, empowering the tracker to adapt to physical deformations without being derailed by corrupted data.

Notably, as shown in Fig. 1, compared to existing trackers including CoReTrack (Dong et al., 2026), MPANet (Liu et al., 2025c), CSTNet (Li et al., 2025), ViPT (Zhu et al., 2023), SDSTrack (Hou et al., 2024), TBSI (Hui et al., 2023), and TATrack (Wang et al., 2024a), our CoManTrack achieves a highly competitive precision on LasHeR (Li et al., 2021) while operating at a real-time speed of 50 FPS.

Our main contributions are as follows:

- (1) We propose CoManTrack, a spatio-temporally decoupled framework that isolates spatial purification from temporal adaptation to halt error propagation and long-term drift.
- (2) We introduce Conflict-Driven Directed Dampening (CD3) to dynamically gate and suppress thermal distractors based on semantic divergence, preserving vital visible structures.
- (3) We design the Tri-Statistic Feature Envelope (TSFE) to capture the holistic structural limits of multi-modal distributions, providing a stable, bounded context that safely guides deep feature aggregation.
- (4) We develop Test-Time Optimal Manifold Adaptation (TOMA), which maintains a prediction-head statistical manifold during inference to filter redundant states and stabilize online adaptation against long-term drift.

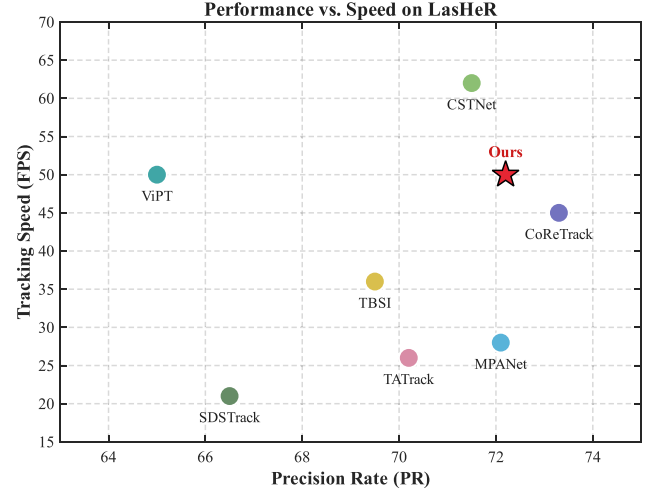


Fig. 1. Precision (PR) and speed (FPS) comparison of our CoManTrack against state-of-the-art trackers on the LasHeR dataset.

2. Related works

2.1. Modality conflict resolution in RGBT tracking

Inevitable conflicts still arise between visible and thermal modalities due to independent physical degradation, challenging their complementary integration in complex tracking environments (Shao et al., 2025a). To reconcile these discrepancies, TBSI (Hui et al., 2023) employs cross-modality attention relying on mutual feature querying to continuously align representations and suppress unshared local background noise. Taking a different route to protect foundational visual knowledge from severe thermal distortion, ViPT (Zhu et al., 2023) treats the infrared stream strictly as an auxiliary prompt injected into a frozen RGB backbone, limiting direct exposure to external interference. Expanding on this paradigm, BAT (Cao et al., 2024) introduces bidirectional adapters generating mutually learning tokens to dynamically recalibrate modality-specific features and filter localized spatial noise before final fusion. Further addressing inaccurate attention allocation during extreme degradation, CAFormer (Xiao et al., 2025) explicitly modulates correlation weights to enforce inter-modality consistency, preventing corrupted regions from dominating the fused representation.

These existing paradigms inherently assume baseline stream reliability, inadvertently allowing aggressive thermal distractors to hijack the shared feature space during extreme degradation. Since mutual querying and consistency-driven modulations fail when global semantics severely diverge, robust localization requires abandoning passive feature reconciliation. Instead, actively evaluating macroscopic semantic divergence motivates a directed dampening strategy that explicitly suppresses high-energy infrared noise and protects fragile visible structures, thoroughly purifying spatial representations prior to temporal integration.

2.2. Spatial feature aggregation and representation

After isolating sensory noise, aggregating purified spatial features into a discriminative representation is critical. In the single-modality domain, foundational one-stream frameworks like OSTRack (Ye et al., 2022) use unified self-attention to compute attention-weighted token summations, flattening spatial dimensions to capture global context.

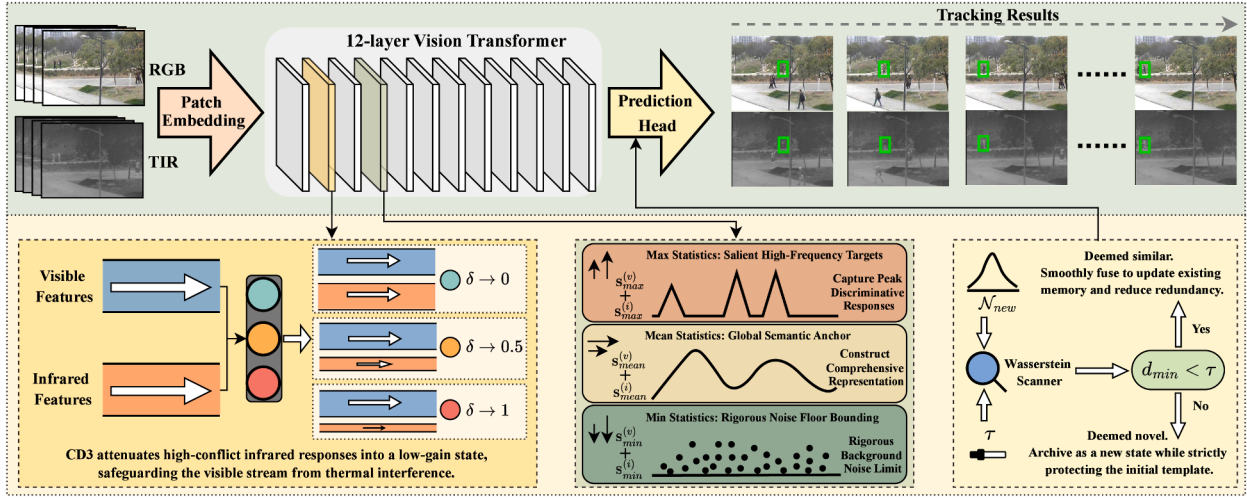


Fig. 2. Overall architecture of the proposed CoManTrack. CD3 and TSFE are embedded in the 2nd and 4th layers of the backbone, respectively, while TOMA is applied during the subsequent testing phase.

Building upon this, recent advancements such as ODTrack (Zheng et al., 2024) employ omni-dimensional attention to enrich these global token relationships, while SPMTrack (Cai et al., 2025) utilizes mixture-of-experts to dynamically route spatial embeddings through specialized pathways. Furthermore, modern generative architectures like ARTrackV2 (Bai et al., 2024) condense visual sequences into compact, task-specific descriptors using autoregressive prompts. Ultimately, despite their architectural diversity, these state-of-the-art pipelines universally rely on mean-centric attention distributions or global average pooling to summarize the target's layout.

However, these aggregation paradigms fundamentally act as low-pass filters, over-smoothing token distributions and inadvertently washing out the localized peaks essential for precise target discrimination. Overcoming this bottleneck necessitates a shift from singular mean-based reductions to a multi-granularity perception paradigm. By synchronously integrating token-level extreme statistics with average responses, this approach effectively bridges global context and local saliency, ensuring the stable retention of fine-grained discriminative features.

2.3. Test-time adaptation

Test-Time Adaptation (TTA) dynamically calibrates models against distribution shifts during inference without accessing the original training data. AdaBN (Li et al., 2016b) pioneers training-free domain alignment by directly replacing the source-domain Batch Normalization statistics with the target-domain mean and variance. To further refine feature representations, Tent (Wang et al., 2020) optimizes the affine transformation parameters of normalization layers via gradient descent by minimizing the Shannon entropy of the model's predictions. Addressing the challenge of long-term continuous adaptation, CoTTA (Wang et al., 2022) employs mean-teacher frameworks and robust data augmentation to enforce consistency regularization, thereby mitigating gradual error accumulation over time. Meanwhile, seeking to prevent catastrophic forgetting and feature collapse during continual adaptation, methods like TCA (Ni et al., 2025) explicitly maintain consistent inter-class topological structures and intra-class compactness to stabilize network representations.

However, applying these generic paradigms to continuous tracking introduces severe vulnerabilities: gradient optimizations cause prohibitive latency, while passive memory queues blindly absorb temporal redundancy. To prevent memory corruption during target stalls, robust tracking requires actively distilling historical states. Consequently, uti-

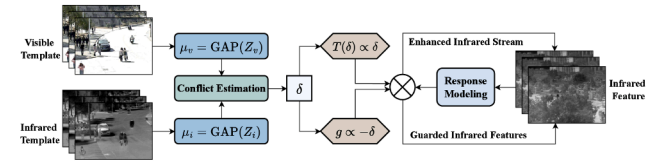


Fig. 3. Architecture of the conflict-driven directed dampening (CD3) module.

lizing geometric metrics to assess frame novelty and construct a well-conditioned pseudo-gradient matrix offers a stable, anti-forgetting adaptation pathway.

3. Method

3.1. Overview

Fig. 2 illustrates the overall architecture of CoManTrack. Built upon a standard 12-layer ViT backbone (Dosovitskiy, 2020), the framework explicitly decouples spatial feature purification from temporal memory adaptation to prevent error propagation and long-term drift.

In the spatial pathway, the 2nd Transformer layer incorporates Conflict-Driven Directed Dampening (CD3) to suppress thermal distractors based on semantic divergence, while the 4th layer applies the Tri-Statistic Feature Envelope (TSFE) to prevent variance explosion during cross-modal aggregation using extreme-statistic prompts. Post-backbone, temporal dynamics are governed exclusively during the test phase by the Test-Time Optimal Manifold Adaptation (TOMA) module. By leveraging the 2-Wasserstein distance to filter redundant historical states, TOMA maintains a well-conditioned memory buffer for robust SVD recovery and stable long-term predictions.

3.2. Conflict-driven directed dampening

To actively prevent aggressive infrared distractors from hijacking the shared feature space, we introduce the Conflict-Driven Directed Dampening (CD3) strategy, whose structure is illustrated in Fig. 3. Given the visible and infrared features X_v, X_i extracted by the backbone, we decouple them into template (Z_v, Z_i) and search region (S_v, S_i) features. To capture the macroscopic semantic representations of each modality, we compute their distribution centers μ_v, μ_i via Global Average Pooling (Lin et al., 2013) and quantify their semantic divergence using a

cosine-distance-based conflict score δ :

$$\mu_m = \frac{1}{L_z} \sum_{j=1}^{L_z} Z_m^{(j)}, \quad m \in \{v, i\}, \quad (1)$$

$$\delta = \frac{1}{2} \left(1 - \frac{\langle \mu_v, \mu_i \rangle}{\|\mu_v\|_2 \|\mu_i\|_2} \right), \quad (2)$$

where L_z denotes the token sequence length of the template features, and $Z_m^{(j)}$ signifies the feature vector of the j th token along the sequence dimension for modality m . To maintain numerical stability during mixed-precision training, the cosine similarity is strictly bounded within $[-1 + \epsilon, 1 - \epsilon]$, with a minimal constant $\epsilon = 10^{-6}$. A scalar $\delta \rightarrow 1$ indicates severe semantic divergence and a highly degraded state.

Upon deriving δ , the network employs the CD3 mechanism for active defense. Instead of symmetric processing, it executes a modality-specific directed dampening strategy. Since infrared degradation (e.g., thermal crossover or background heat sources) typically generates aggressive, high-energy false-positive peaks that easily hijack cross-modal attention, CD3 is explicitly deployed on the infrared stream X_i to act as a thermal dampening gate. It should be noted that this directed design does not assume universal visible superiority. CD3 is a lightweight early-layer conflict-control module for active thermal-distractor cases, while useful infrared cues can still be retained by the subsequent TSFE module through multi-granularity statistical enrichment and adaptive routing.

We first compute its nonlinear activation base $Y = \phi(X_i)$ and constrained channel mean μ_Y to obtain the basic discriminative response B :

$$\mu_Y = \max \left(\frac{1}{N} \sum Y, \mu_{clamp} \right), \quad (3)$$

$$B = \frac{Y^2}{\mu_Y}, \quad (4)$$

where ϕ denotes the nonlinear activation function, N represents the total number of spatial tokens, and μ_{clamp} is a predefined lower-bound constant to prevent division by zero.

To dynamically suppress extreme thermal noise, CD3 introduces a temperature scaling mapping $T(\delta)$ and a phase-transition gating function $g(\delta)$, both strictly driven by δ :

$$T(\delta) = t_{min} + (t_{max} - t_{min}) \cdot \delta, \quad (5)$$

$$g(\delta) = \frac{1}{1 + \exp(-k \cdot (d_0 - \delta))}, \quad (6)$$

where $t_{min} = 1.0$ and $t_{max} = 3.0$ define the lower and upper bounds of the temperature scaling, $k = 8.0$ controls the steepness of the gating curve, and $d_0 = 0.3$ specifies the center of the soft conflict transition. These values are empirically selected and further evaluated through the hyperparameter sensitivity analysis. Since $T(\delta)$ increases monotonically while $g(\delta)$ decreases monotonically with the conflict score δ , CD3 progressively shifts toward a conservative attenuation state without introducing an abrupt hard threshold.

Subsequently, the final feature modulation weight W and the enhanced infrared output $\hat{X}_i$ are formulated as:

$$W = \frac{1 - g(\delta)}{2} + g(\delta) \cdot \phi \left(\frac{B}{T(\delta)} \right), \quad (7)$$

$$\hat{X}_i = X_i \odot W, \quad (8)$$

where $\odot$ denotes the element-wise Hadamard product.

Fundamentally, this directed formulation functions as a bounded dynamic dampening gate rather than a hard blocking mask. As the modality conflict δ exceeds the safety threshold d_0 , $g(\delta)$ decays rapidly towards zero, while the climbing temperature $T(\delta)$ forces the activation $\phi(B/T(\delta))$ into an attenuated state. Consequently, the modulation weight W shifts toward a conservative low-gain branch instead of approaching zero. This mechanism dampens aggressive infrared responses while preserving residual thermal cues, thereby safeguarding visible features and incubating a highly reliable shared search stream.

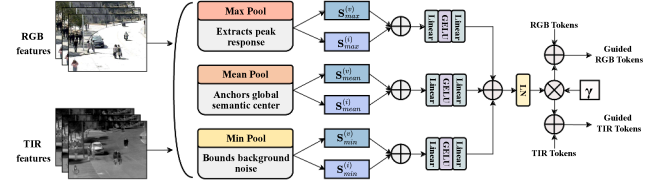


Fig. 4. Architecture of the tri-statistic feature envelope (TSFE) module for multi-granularity feature aggregation.

3.3. Multi-granularity perception via tri-statistic feature envelope

Traditional single-statistic pooling (e.g., mean-pooling) inevitably leads to over-smoothed representations, masking high-frequency peak target responses and background boundaries. To overcome this representational bottleneck and comprehensively capture multi-granularity spatial contexts, we design the Tri-Statistic Feature Envelope (TSFE) module, whose detailed structure is illustrated in Fig. 4. At the token level, this module synchronously decouples three orthogonal pooling states—minimum, mean, and maximum—constructing a comprehensive multi-granularity feature representation.

Given the modality-specific features F_m , we initially cast them to FP32 to prevent precision overflow and compute the mean, maximum, and minimum statistics along the spatial dimension:

$$S_{mean}^{(m)} = \frac{1}{N} \sum_{j=1}^N F_m^{(j)}, \quad (9)$$

$$S_{max}^{(m)} = \max_{1 \leq j \leq N} F_m^{(j)}, \quad (10)$$

$$S_{min}^{(m)} = \min_{1 \leq j \leq N} F_m^{(j)}, \quad (11)$$

where N denotes the total sequence length, and $F_m^{(j)}$ signifies the j th token's feature vector. Mathematically, these operators serve complementary functions: S_{mean} anchors the global semantic center, S_{max} extracts peak discriminative responses of high-frequency salient features, and S_{min} rigorously bounds the lower limit of background noise.

Instead of relying on naive element-wise addition for cross-modal alignment, which risks injecting noise from degraded modalities (e.g., infrared distractors or overexposed visible regions), we dynamically assign fusion weights for each pooling operator $op \in \{min, mean, max\}$ through a modality gating operation:

$$G_{op} = \text{Sigmoid} \left(W_{gate} [S_{op}^{(v)}; S_{op}^{(i)}] \right), \quad (12)$$

$$S_{op}^{(fused)} = G_{op} \odot S_{op}^{(v)} + (1 - G_{op}) \odot S_{op}^{(i)}, \quad (13)$$

where $[:]$ denotes channel-wise concatenation, and W_{gate} represents the learnable parameters.

These decoupled and securely fused features are then mapped into a shared semantic space via a non-linear projection f_{MLP} (a fully connected layer followed by a GELU activation). To aggregate them into a unified multi-granularity prior U_{global} , we adaptively allocate attention to different granularities based on the current tracking context (e.g., emphasizing S_{max} during occlusion to identify peak responses, or S_{min} and S_{mean} to suppress severe background clutter). Rather than rigidly summing the statistics, the features $H_{op} = f_{MLP}(S_{op}^{(fused)})$ are dynamically routed using weights α , generated as follows:

$$[\alpha_{min}, \alpha_{mean}, \alpha_{max}] = \text{Softmax} \left(W_{router} [H_{min}; H_{mean}; H_{max}] \right), \quad (14)$$

$$U_{global} = \sum_{op} \alpha_{op} H_{op}, \quad (15)$$

where W_{router} maps the concatenated multi-granularity features into a three-dimensional weight distribution.

Directly incorporating the S_{max} operator into a deep Transformer risks geometrically amplifying feature variance, precipitating gradient

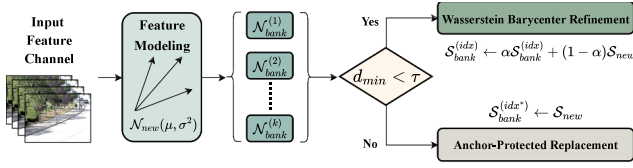


Fig. 5. TOMA's distance-driven adaptation and update mechanism.

collapse during mixed-precision training. To circumvent this numerical instability, we apply standard Layer Normalization to stabilize the fused representation. Specifically, U_{global} is standardized to strictly penalize variance explosion, yielding the final multi-granularity prompt P_{global} through an affine transformation:

$$\hat{U}_{global} = \frac{U_{global} - \mathbb{E}[U_{global}]}{\sqrt{\text{Var}[U_{global}] + \epsilon}}, \quad (16)$$

$$P_{global} = \gamma_{global} \odot \hat{U}_{global} + \beta_{global}, \quad (17)$$

where $\epsilon = 10^{-6}$ ensures numerical stability, while γ_{global} and β_{global} denote the standard learnable scaling and translation factors of LN.

Residually added to the primary feature stream, P_{global} serves as a multi-granularity prior for the tracking framework. Supported by this variance stabilization, the decoupled routing design overcomes the over-smoothing limitations of standard mean-pooling, effectively maintaining high-frequency spatial details and peak target responses during multimodal integration.

3.4. Test-time optimal manifold adaptation

Online tracking paradigms frequently suffer from severe temporal redundancy and catastrophic forgetting as the target undergoes continuous appearance variations. To mitigate these degradation issues and maintain a highly discriminative historical context, we develop the Test-Time Optimal Manifold Adaptation (TOMA) module, whose overall framework is illustrated in Fig. 5. By leveraging optimal transport theory, TOMA actively distills a diverse, non-redundant state space within a statistical manifold bank.

In the implementation, TOMA performs lightweight test-time adaptation only on the BatchNorm statistics of the convolutional prediction head, while the ViT backbone and all learnable network weights remain frozen during inference. Importantly, the manifold bank is directly coupled with the tracking objective rather than serving as an auxiliary frame memory. Each stored state represents the BatchNorm distribution of target-conditioned prediction-head activations, which directly determine the classification response, center offset, and bounding-box regression. Since the ViT backbone does not contain BatchNorm layers, the statistical states used by TOMA are collected from the prediction-head BatchNorm layers. Specifically, TOMA adapts the running mean and running variance by modeling the prediction-head activation distribution as a diagonal Gaussian $\mathcal{N}(\mu, \sigma^2)$. For a newly observed distribution $\mathcal{N}_{new}$ at frame t , we quantify its geometric divergence from the stored distributions $\mathcal{N}_{bank}$ using the 2-Wasserstein distance:

$$W_2^2(\mathcal{N}_{new}, \mathcal{N}_{bank}) = \|\mu_{new} - \mu_{bank}\|_2^2 + \|\sigma_{new} - \sigma_{bank}\|_2^2, \quad (18)$$

where $\sigma = \sqrt{\text{Var} + \epsilon}$ utilizes the channel variance, and $\epsilon = 10^{-6}$ ensures numerical stability.

To evaluate the informational value of the incoming observation during adaptation, we compute its minimum transport cost against all stored states:

$$d_{min} = \min_{idx} W_2(\mathcal{N}_{new}, \mathcal{N}_{bank}^{(idx)}). \quad (19)$$

The update mechanism depends on a predefined distance threshold τ . If $d_{min} < \tau$, the observation is considered redundant and is compressed via Wasserstein barycenter refinement. Defining the statistical

state vector as $S = [\mu, \sigma]^T$, we formulate the Fréchet functional $F(S)$ as the weighted transport cost:

$$F(S) = \alpha W_2^2(S, S_{bank}^{(idx)}) + (1 - \alpha) W_2^2(S, S_{new}). \quad (20)$$

The updated state minimizes this objective:

$$S_{bank}^{(idx)} \leftarrow \arg \min_S F(S). \quad (21)$$

Given the diagonal Gaussian formulation, this admits a closed-form geometric update:

$$S_{bank}^{(idx)} \leftarrow \alpha S_{bank}^{(idx)} + (1 - \alpha) S_{new}, \quad (22)$$

where α is the refinement momentum. This continuous refinement prevents memory pollution while retaining historical context during the test-time phase.

Conversely, if $d_{min} \geq \tau$, the observation represents a novel manifold state. If the bank reaches its maximum capacity K , we apply an anchor-protected replacement strategy: the new state overwrites the non-initial stored state exhibiting the minimal W_2 distance. This explicitly protects the initial ground-truth template from being discarded.

Let $S_0 = [\mu_0, \sigma_0]^T$ denote the base BatchNorm statistical state, and let $S_{bank}^{(k)} = [\mu_{bank}^{(k)}, \sigma_{bank}^{(k)}]^T$ denote the k th distributionally selected state in the manifold bank. Different from a chronological memory queue, these states are maintained according to Wasserstein novelty, where redundant observations are compressed and novel target appearances are preserved. Based on this compact manifold, we construct the statistical difference matrix:

$$\Delta_{stat} = [S_0 - S_{bank}^{(1)}, S_0 - S_{bank}^{(2)}, \dots, S_0 - S_{bank}^{(K)}]. \quad (23)$$

We then perform SVD on the covariance matrix constructed from the centered statistical differences in Δ_{stat} to estimate the dominant distribution-shift directions within the maintained target manifold. The resulting singular vectors define the principal recovery directions of the BatchNorm statistics, while the corresponding singular values weight the contribution of each direction. The weighted recovery direction is directly applied to recover the mean and standard deviation of the prediction-head BatchNorm layers using separate adaptation step sizes, with the recovered standard deviation squared to obtain the corresponding running variance. The singular values further regulate the momentum of subsequent statistical updates. The recovered statistics are immediately used by the prediction head for target classification and bounding-box regression, while the ViT backbone and all learnable network weights remain frozen. Since Δ_{stat} is constructed from Wasserstein-filtered states, the recovery is driven by diverse target variations rather than temporally repetitive observations.

3.5. Loss function

The output token sequence from the backbone is reshaped into a 2D spatial feature map and fed into a fully convolutional prediction head. This head generates the target classification score map, the center offset, and the normalized bounding box coordinates.

The overall network is optimized end-to-end using a combination of classification and regression objectives, formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_1 + \lambda_{iou} \mathcal{L}_{iou}. \quad (24)$$

Here, $\mathcal{L}_{cls}$ represents the weighted focal loss for target classification, while the standard L_1 loss $\mathcal{L}_1$ and the generalized Intersection over Union (GIoU) loss $\mathcal{L}_{iou}$ supervise the bounding box regression. The coefficients λ_1 and λ_{iou} are trade-off parameters, empirically set to 5.0 and 2.0, respectively, to balance the gradient magnitudes.

4. Experiments

4.1. Implementation details

We implement the proposed CoManTrack using PyTorch 2.7.1 (Paszke et al., 2019) and Python 3.9 on an Ubuntu 20.04

Table 1
Quantitative results on the GTOT dataset. Best in underline, second-best in **bold**.

Methods	Pub. & Year	GTOT	
		PR	SR
APFNet (Xiao et al., 2022)	AAAI 2022	90.5	73.7
DMCNet (Lu et al., 2022)	TNNLS 2022	90.9	73.3
CMD (Zhang et al., 2023)	CVPR 2023	89.2	73.4
QAT (Liu et al., 2023)	ACM MM 2023	91.5	75.5
BAT (Cao et al., 2024)	AAAI 2024	90.9	76.3
CAT + (Liu et al., 2024)	TIP 2024	91.5	73.3
FFE-BMFF (Zhang et al., 2025b)	Neurocomp. 2024	92.4	77.0
AFTER (Lu et al., 2025a)	TIP 2025	91.6	<u>78.5</u>
CAFormer (Xiao et al., 2025)	AAAI 2025	91.8	76.9
MAT (Wang et al., 2025)	Inf. Fusion 2025	91.0	76.5
MGNet (Zhang et al., 2025c)	Neural Netw. 2025	90.0	75.0
MPANet (Liu et al., 2025c)	TMM 2025	92.3	76.8
SAFNet (Liu et al., 2025d)	Neurocomp. 2025	92.1	76.4
TAAT (Tang et al., 2025a)	PRL 2025	88.9	71.7
TCM (Zhao et al., 2026)	IVC 2026	89.6	75.5
Ours	2026	<u>92.6</u>	76.9

system. All model training and evaluation are executed on a single 32GB NVIDIA RTX 5090D GPU. Specifically, the model is optimized on the LasHeR training split and evaluated on its corresponding testing set.

During the 16-epoch training phase, each epoch processes 60,000 randomly sampled image pairs organized in mini-batches of 32. Input dimensions are standardized to 128×128 for templates and 256×256 for search areas. Model parameters are updated via the AdamW optimizer (Loshchilov & Hutter, 2017). The base learning rate is set to 1×10^{-4} (with the backbone scaled to 1×10^{-5}) and decays by an order of magnitude at epoch 10. To prevent overfitting and maintain training stability, we adopt a weight decay of 1×10^{-4} , a DropPath rate of 0.1, and a gradient clipping norm of 0.1.

4.2. Evaluation on public datasets

To comprehensively evaluate the performance and generalization ability of CoManTrack, we conduct extensive experiments on three widely used RGBT tracking benchmarks, namely GTOT (Li et al., 2016a), RGBT234 (Li et al., 2019), and LasHeR. In alignment with standard practices within the visual tracking community, the quantitative evaluation is primarily based on two metrics: Precision Rate (PR) for measuring center location error and Success Rate (SR) for assessing bounding box overlap. In addition to these standard metrics, we specifically report the Normalized Precision Rate (NPR) when evaluating on the LasHeR benchmark, which provides additional insights into the tracker’s precision under varying target scales.

4.2.1. GTOT

The GTOT dataset contains 50 aligned RGB and thermal video sequences designed for short-term tracking evaluation. As shown in Table 1, our tracker achieves the highest Precision Rate (PR) of 92.6% among 15 recent state-of-the-art methods, slightly outperforming FFE-BMFF (92.4%). In terms of Success Rate (SR), the method secures a competitive 76.9%, tying with CAFormer and closely following FFE-BMFF (77.0%). These metrics confirm its reliable target localization in standard RGBT scenarios.

As depicted by the radar chart in Fig. 6, we further analyze the tracking robustness of our method against five other trackers, namely ECO (Danelljan et al., 2017), CAT (Li et al., 2020), SGT (Li et al., 2017), AFTER (Lu et al., 2025a), and CAFormer (Xiao et al., 2025), across seven challenging attributes on the GTOT dataset: OCC, LSV, FM, LI, TC, SO, and DEF. Our approach demonstrates exceptional center localization capability, achieving the highest PR in the OCC (90.7%) and TC (92.6%)

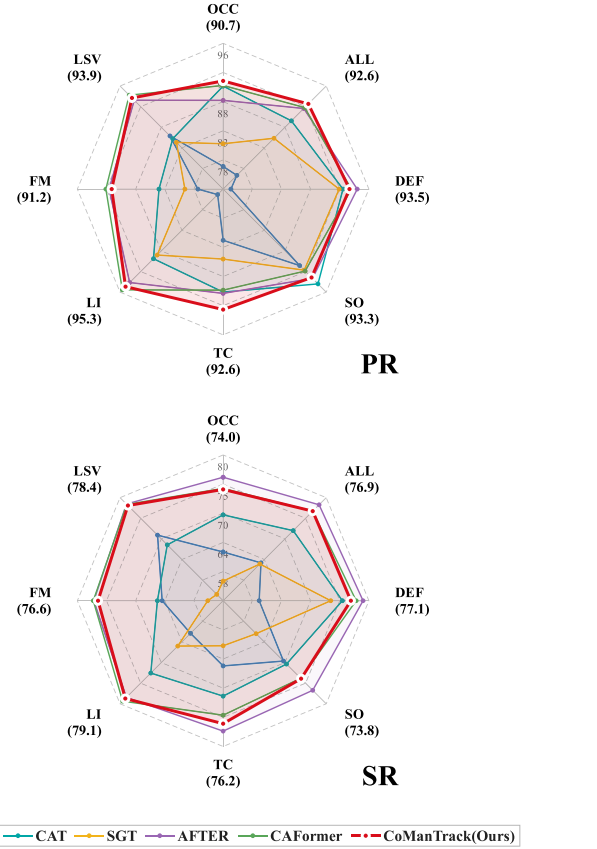


Fig. 6. Radar charts detailing the attribute-wise PR and SR performance on the GTOT dataset. The annotated numerical values denote the specific scores achieved by our CoManTrack.

categories. Furthermore, it delivers highly competitive PR scores in LI (95.3%) and LSV (93.9%), closely following CAFormer. While trackers like AFTER show strong SR in specific scenarios, our method maintains a highly stable balance across all attributes, ultimately yielding the highest overall PR of 92.6% and a robust overall SR of 76.9%.

This balanced robustness across diverse challenges fundamentally stems from our spatio-temporally decoupled architecture. The strong performance under TC and LI is driven by the directed dampening mechanism, which actively suppresses severe thermal noise to prevent distractors from polluting the shared feature space. For spatial distortions like OCC and LSV, the network extracts normalized extreme statistics instead of relying on standard mean-pooling. This preserves fine-grained structural cues, ensuring tight target anchoring even under severe partial occlusion. Furthermore, the overall stability against dynamic changes, such as DEF and FM, is maintained by our optimal transport-guided temporal adaptation. By leveraging optimal transport costs to filter redundant observations, the method curates a well-conditioned memory matrix, effectively halting the propagation of localized spatial errors into long-term tracking drift.

4.2.2. RGBT234

To evaluate performance in more complex real-world scenarios, we utilize the large-scale RGBT234 benchmark. Comprising 234 aligned sequence pairs with rich attribute annotations, it provides a highly rigorous testing environment. As detailed in Table 2, our method achieves a top Precision Rate (PR) of 87.6%, outperforming strong baselines like MPANet. For Success Rate (SR), it secures a competitive 65.3%, ranking second only to CTGTrack and surpassing CAFM. These metrics confirm the tracker’s robust center localization and bounding box estimation in cluttered environments.

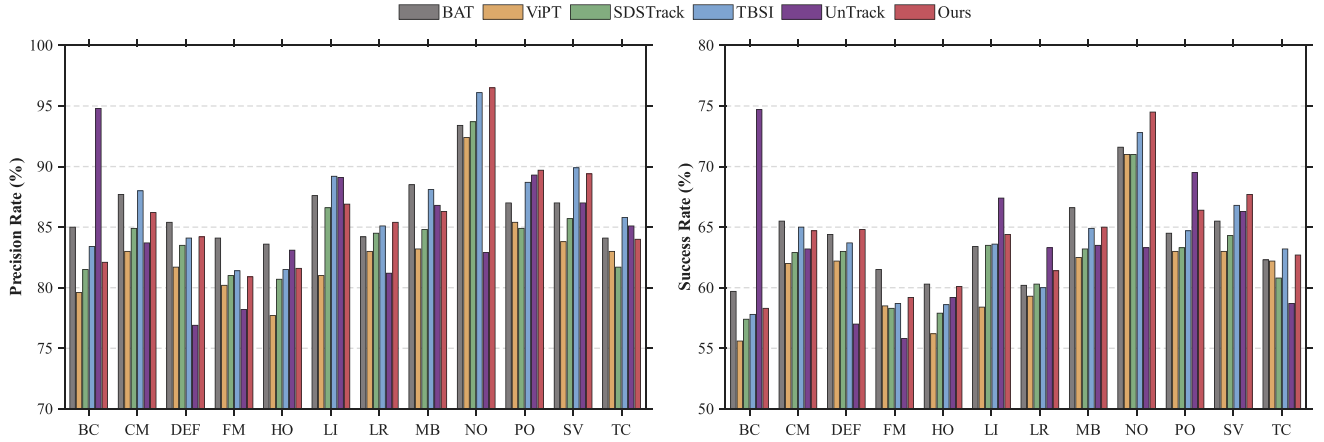


Fig. 7. Attribute-based comparison of precision rate and success rate against state-of-the-art trackers on the RGBT234 dataset.

Table 2

Quantitative results on the RGBT234 dataset. Best in underline, second-best in **bold**.

Methods	Pub. & Year	RGBT234	
		PR	SR
TBSI (Hui et al., 2023)	CVPR 2023	87.1	63.7
ViPT (Zhu et al., 2023)	CVPR 2023	83.5	61.7
BAT (Cao et al., 2024)	AAAI 2024	86.8	64.1
FFE-BMFF (Zhang et al., 2025b)	Neurocomp. 2024	85.1	62.8
OneTracker (Hong et al., 2024)	CVPR 2024	85.7	64.2
SDSTrack (Hou et al., 2024)	CVPR 2024	84.8	62.5
SiamTFA (Zhang et al., 2024a)	TITS 2024	82.2	59.2
CAFM (Zhang et al., 2025a)	SPL 2025	86.8	64.6
MCFTNet (Liu et al., 2025b)	ASC 2025	84.2	60.7
MCIT (Qin et al., 2025)	Neurocomp. 2025	83.1	59.5
MGNet (Zhang et al., 2025c)	Neural Netw. 2025	85.2	63.7
MPANet (Liu et al., 2025c)	TMM 2025	87.4	64.6
PTMA (Luo et al., 2025)	TIM 2025	85.8	64.1
CTGTrack (Wei et al., 2026)	PR 2026	87.2	<u>65.9</u>
TCM (Zhao et al., 2026)	IVC 2026	86.2	64.0
Ours	2026	<u>87.6</u>	65.3

Table 3

Quantitative results on the LasHeR dataset. Best in underline, second-best in **bold**.

Methods	Pub. & Year	LasHeR		
		PR	NPR	SR
TBSI (Hui et al., 2023)	CVPR 2023	69.2	65.7	55.6
ViPT (Zhu et al., 2023)	CVPR 2023	65.1	–	52.5
AMNet (Zhang et al., 2024b)	TCSVT 2024	70.2	66.0	55.9
BAT (Cao et al., 2024)	AAAI 2024	70.2	66.4	56.3
OneTracker (Hong et al., 2024)	CVPR 2024	67.2	–	53.8
SDSTrack (Hou et al., 2024)	CVPR 2024	66.5	–	53.1
AFTER (Lu et al., 2025a)	TIP 2025	70.3	65.8	55.1
CAFormer (Xiao et al., 2025)	AAAI 2025	70.0	66.1	55.6
MGNet (Zhang et al., 2025c)	NN 2025	70.0	65.9	56.0
MPANet (Liu et al., 2025c)	TMM 2025	72.1	–	57.3
RAMR (Gao et al., 2025)	ESWA 2025	71.9	68.5	57.2
SMGNet (Liu et al., 2025a)	PR 2025	72.0	<u>68.0</u>	<u>57.4</u>
TUFNet (Liu et al., 2025e)	KBS 2025	70.8	67.4	55.7
CTGTrack (Wei et al., 2026)	PR 2026	69.5	–	55.0
TCM (Zhao et al., 2026)	IVC 2026	71.0	67.0	56.5
Ours	2026	<u>72.2</u>	68.2	<u>57.4</u>

Beyond the overall metrics, a detailed attribute-based analysis in Fig. 7 further investigates the specific strengths of our tracker across twelve distinct challenge factors. Specifically, to comprehensively evaluate its robustness, we compare our approach against five recent state-of-the-art trackers—BAT (Cao et al., 2024), ViPT (Zhu et al., 2023), SDSTrack (Hou et al., 2024), TBSI (Hui et al., 2023), and UnTrack (Wu et al., 2024)—on the RGBT234 dataset. The evaluation indicates that our method exhibits particular resilience to target appearance changes and structural variations. Specifically, it delivers the best PR in LR, PO, and NO scenarios, while simultaneously leading the SR metric under DEF, SV, and NO. Even in traditionally demanding conditions such as HO, LI, and FM, the proposed method consistently secures top-tier rankings against strong competitors. Rather than overfitting to a narrow set of simple scenarios, this comprehensive and well-rounded robustness across various challenging attributes directly contributes to its leading overall precision and tracking stability.

The overall stability of the model clearly shows the advantage of our spatio-temporally decoupled design. It handles structural and appearance changes remarkably well, especially in LR, PO, and SV scenarios. We attribute this to the use of normalized extreme statistics. Standard pooling tends to over-smooth features, but our method retains the fine structural details. These details are vital for accurately estimating the target's scale and boundaries when it is visually degraded or partly covered. In tougher situations like LI and HO, the framework maintains a reliable shared representation by attenuating high-conflict infrared

responses when necessary and preserving complementary cross-modal cues through subsequent statistical fusion. Additionally, our optimal transport-guided temporal adaptation is the main reason we achieve such high success rates with DEF and handle FM so effectively. By constantly clearing out redundant past states, the network keeps its memory efficient. This allows it to adapt smoothly when the target changes shape without letting small temporal errors build up.

4.2.3. LasHer

The LasHeR benchmark holds particular significance in our experiments, as its massive scale of 1224 sequence pairs provides the diverse conditions necessary for training our model. To verify how well the network generalizes from this extensive training, we thoroughly evaluate it on the official LasHeR testing set. As summarized in Table 3 and visually illustrated in Fig. 8, our method demonstrates highly competitive and balanced results, setting a new state-of-the-art PR of 72.2%. It also secures a strong second place in NPR (68.2%) just behind RAMR, and shares the top rank in SR (57.4%) alongside SMGNet. These metrics validate the effectiveness of our approach in dealing with extreme environmental variations and massive-scale sequences.

A more detailed attribute analysis in Table 4 provides further insights into the tracker's behavior. Across the 19 challenge factors, the method establishes a dominant lead in precision, ranking first in PR for 14 categories (e.g., AIV, DEF, HO, and MB). For bounding box accuracy (SR), it ranks first in eight attributes, showing particular strength in handling

Table 4
Attribute-based comparison with state-of-the-art trackers on the LasHeR dataset. underline, **bold** and *green* fonts indicate the best, second-best, and third-best results, respectively.

PR	ARC	SV	FM	OV	FL	TC	CM	SA	BC	DEF	LR	AIV	HI	LI	MB	HO	TO	PO	NO	ALL
AFTER (Lu et al., 2025a)	65.1	70.6	68.8	69.9	57.9	63.2	69.1	61.0	67.0	72.4	63.1	58.7	78.0	60.6	65.0	57.8	62.4	67.4	91.0	70.3
BAT (Cao et al., 2024)	63.1	69.7	68.5	76.8	62.0	62.7	68.0	61.3	67.7	72.2	62.7	51.4	75.3	60.4	62.4	56.5	64.1	67.5	90.2	70.2
CAFormer (Xiao et al., 2025)	63.3	69.5	68.8	72.8	59.1	62.3	68.3	61.8	70.0	73.9	61.5	56.4	73.3	60.3	63.3	59.0	64.3	67.2	89.6	70.0
GMMT (Tang et al., 2024b)	64.7	70.6	69.2	67.5	63.6	63.1	69.0	61.9	68.8	70.8	63.7	53.0	75.2	61.9	64.0	57.6	64.1	68.0	90.9	70.7
MoETrack (Tang et al., 2025b)	64.5	71.7	70.7	71.1	63.1	64.4	70.4	63.3	71.0	73.3	64.5	54.0	76.6	62.2	64.1	57.2	65.5	69.5	91.0	72.1
Ours	65.1	71.9	70.7	75.1	62.0	65.1	70.7	64.0	71.0	74.6	64.0	60.3	74.9	61.3	65.4	60.5	67.0	69.6	91.6	72.2
SR	ARC	SV	FM	OV	FL	TC	CM	SA	BC	DEF	LR	AIV	HI	LI	MB	HO	TO	PO	NO	ALL
AFTER (Lu et al., 2025a)	52.0	55.3	54.3	59.9	43.2	49.1	53.5	48.0	52.3	58.3	45.0	49.6	59.3	48.0	49.5	50.4	48.7	52.8	72.2	55.1
BAT (Cao et al., 2024)	51.9	55.9	55.1	66.0	49.0	50.1	54.4	49.2	53.9	58.5	46.2	45.3	59.6	48.2	49.6	51.0	51.1	54.0	73.3	56.3
CAFormer (Xiao et al., 2025)	51.8	55.3	55.1	61.7	46.3	49.4	54.2	49.3	55.2	60.0	44.8	48.8	57.5	48.3	49.8	52.4	51.0	53.5	71.9	55.6
GMMT (Tang et al., 2024b)	52.8	56.7	55.6	58.8	49.8	50.5	54.7	50.2	54.7	58.0	47.3	46.4	59.3	49.5	50.5	51.2	50.9	54.4	74.3	56.6
MoETrack (Tang et al., 2025b)	53.0	57.7	56.9	61.5	50.3	51.6	56.2	51.0	56.4	59.7	47.9	47.6	60.6	49.8	50.6	51.3	52.0	55.6	74.5	57.8
Ours	52.9	57.2	56.6	64.1	48.8	51.6	55.9	51.2	56.5	61.0	46.9	52.5	59.4	49.4	51.5	53.4	52.7	55.2	74.0	57.4

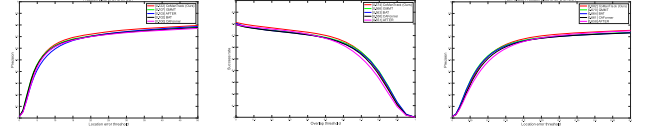


Fig. 8. Visualization results of PR, SR, and NPR on the LasHeR dataset.

TC, SA, and BC. Unlike methods that excel only in simple scenes, our approach maintains high accuracy even when targets face severe deformation or occlusion, proving its strong generalization capability across highly variable environments.

The ability to generalize across such a massive dataset underscores the practical value of decoupling spatial purification from temporal adaptation. The tracker's clear advantage under AIV and TC is driven directly by the directed dampening mechanism. When active thermal distractors cause substantial cross-modal conflict, CD3 attenuates the corresponding infrared responses. Under visible degradation, robustness mainly benefits from the complementary infrared cues and the adaptive statistical fusion of TSFE. For structural and contextual challenges like HO, BC, and SA, relying on extreme token statistics becomes critical. Because this approach bypasses the over-smoothing typical of average pooling, the network retains the sharp boundary cues needed to separate the true target from complex distractors. Finally, the robust performance against DEF and MB validates our optimal transport-guided memory update. Driven by an optimal transport perspective that rigorously filters out redundant historical states, the memory bank remains well-conditioned, successfully breaking the cycle of error accumulation during rapid physical changes.

4.3. Visualization analysis

4.3.1. RGBT234

The visualized tracking results are presented in Fig. 9, comparing CoManTrack against five representative rivals on the challenging *womancross*, *mancross1*, and *threeman2* sequences. In these scenarios, CoManTrack achieves more stable target association and tighter bounding boxes despite dense crowds, severe occlusion, and background clutter. In contrast, competing methods frequently drift or over-expand their boxes once the dominant modality degrades.

In the *womancross* sequence, the target navigates through a dense gathering of pedestrians. While other trackers easily drift to similar distractors within the crowd, CoManTrack successfully locks onto the specific target and maintains this precise lock even after the target crosses the street. The *mancross1* sequence presents a severe occlusion challenge where the target is completely blocked by a passing bus. Many baseline methods lose the subject entirely or incorrectly anchor to the vehicle; in contrast, our method robustly handles the occlusion and instantly recovers the tracking box once the bus passes. In *threeman2*, the target experiences partial occlusion while walking behind a tree. CoManTrack leverages its robust spatial-temporal features to stably distinguish the target from the obstacle, completely avoiding the severe box drift and scale failures seen in competing trackers.

4.3.2. LasHeR

To provide qualitative insights on the LasHeR dataset, tracking results are visualized across sampled frames from the *right blk boy 4386*, *white skirt girl coming from goal*, and *1st col 4th boy* sequences. As depicted in Fig. 10, CoManTrack is compared against five competing methods. In these challenging scenarios, our approach demonstrates superior reliability by maintaining tighter bounding boxes and completely avoiding the severe spatial drift frequently observed in competing methods.

In the *rightblkboy4386* sequence, the target experiences extreme backlighting and is heavily occluded by structural pillars. While most competing trackers lose the target entirely during the occlusion or fail to

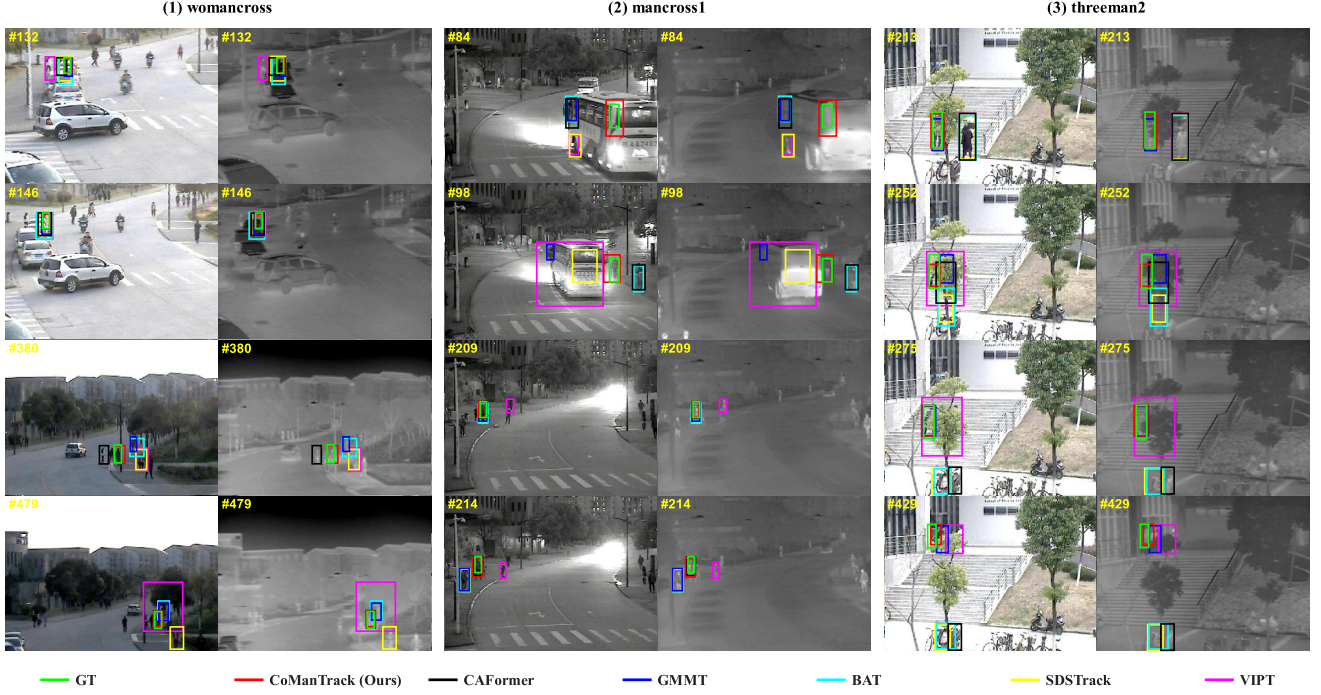


Fig. 9. Visual comparison of our CoManTrack with five other trackers on the *womancross*, *mancros1*, and *threeman2* sequences from the RGBT234 dataset.

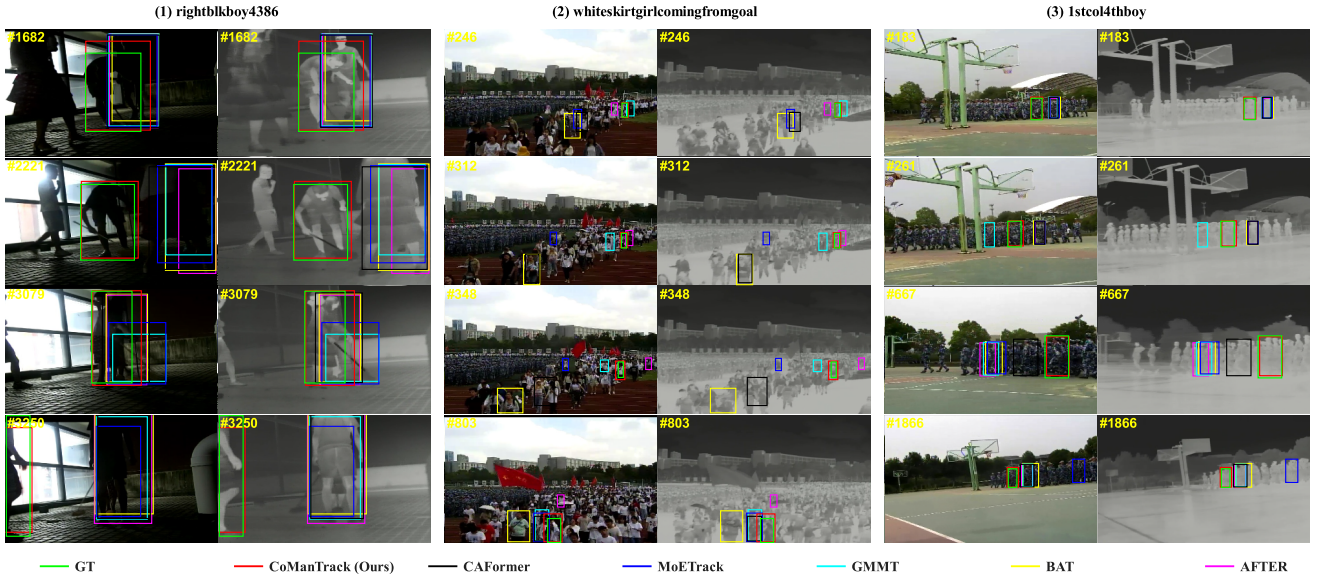


Fig. 10. Visual comparison of our CoManTrack with five other trackers on the *rightblkboy4386*, *whiteskirtgirlcomingfromgoal*, and *1stcol4thboy* sequences from the LasHeR dataset.

adapt to the drastic illumination contrast, CoManTrack retains complementary infrared evidence and robust multi-granularity cues despite the severe degradation of visible observations. The *whiteskirtgirlcomingfromgoal* sequence features a target navigating through a massive, dense crowd of students with nearly identical appearances. In this scenario, baseline methods frequently drift to visually similar distractors in the background; in contrast, our tracker securely isolates the intended subject without being misled. Finally, the *1stcol4thboy* sequence presents a highly challenging formation of identically dressed individuals on a court, causing continuous semantic ambiguity. While other trackers easily confuse the target with adjacent neighbors, CoManTrack leverages its decoupled spatial-temporal representations to effectively distinguish the specific target from the homogeneous group, completely avoiding box drift.

4.4. Ablation study

4.4.1. Effectiveness of proposed components

To evaluate the individual and complementary contributions of CD3, TSFE, and TOMA, we conduct a complete component-wise ablation study on LasHeR and RGBT234, covering a module-free baseline, single-module variants, pairwise combinations, and the complete model. As reported in Table 5, all three components provide overall improvements over the baseline, although their effects vary across metrics. CD3 mainly improves feature reliability through early conflict regulation, TSFE enhances target representation through multi-granularity statistical prompting, and TOMA achieves the strongest standalone improvement by adapting prediction statistics during inference.

Table 5
Component-wise ablation study of CoManTrack on LasHeR and RGBT234.

Components			LasHeR			RGBT234	
CD3	TSFE	TOMA	PR	NPR	SR	PR	SR
–	–	–	69.8	65.8	55.6	85.3	63.7
✓	–	–	70.4	66.2	55.8	85.9	64.0
–	✓	–	70.6	66.0	56.2	86.1	63.8
–	–	✓	71.1	67.2	56.5	86.6	64.5
✓	✓	–	71.2	67.1	56.7	86.3	64.4
✓	–	✓	71.9	67.9	57.2	87.3	65.0
–	✓	✓	71.5	67.7	57.1	87.1	64.9
✓	✓	✓	72.2	68.2	57.4	87.6	65.3

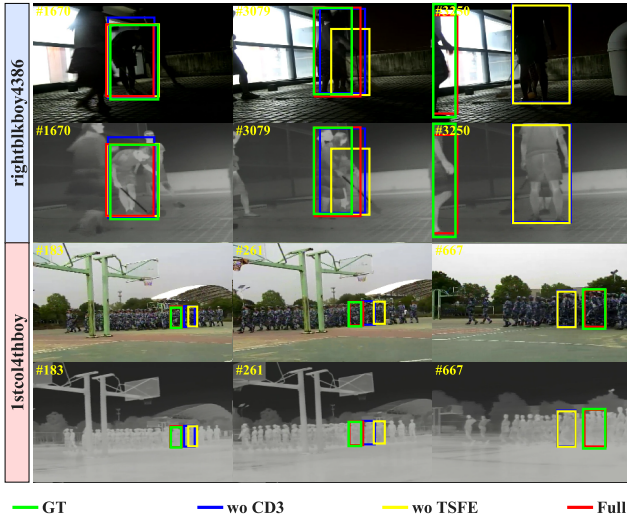


Fig. 11. Qualitative comparison of our full framework and its ablated variants on the LasHeR dataset.

Table 6
Ablation study of the TOMA module at test time.

Variants	LasHeR			RGBT234		FPS
	PR	NPR	SR	PR	SR	
Non-adaptive Baseline	71.2	67.1	56.7	86.3	64.4	65
+ AdaBN (Li et al., 2016b)	71.6	67.5	57.0	86.8	64.9	65
+ Tent (Wang et al., 2020)	71.7	67.5	56.8	87.1	65.0	68
+ EATA (Niu et al., 2022)	71.5	67.7	56.9	87.1	64.8	67
+ TCA (Ni et al., 2025)	71.9	68.0	57.3	87.4	65.3	62
+ TOMA	72.2	68.2	57.4	87.6	65.3	50

The pairwise results further support the functional coupling of the three stages rather than a simple accumulation of independent modules. Specifically, CD3+TSFE validates the ordered spatial pathway, where conflict regulation precedes statistical enrichment, while CD3+TOMA and TSFE+TOMA demonstrate the complementarity between reliable spatial representations and temporal adaptation. Integrating all three components produces the best results, reaching 72.2%/68.2%/57.4% in PR/NPR/SR on LasHeR and 87.6%/65.3% in PR/SR on RGBT234. Compared with the module-free baseline, the complete model improves these metrics by 2.4, 2.4, 1.8, 2.3, and 1.6 percentage points, respectively. Consistent with the quantitative results, Fig. 11 shows that removing individual components makes the tracker more susceptible to distracting responses and localization drift, whereas the complete model maintains more accurate and stable target localization.

Table 7
Internal ablation study of TOMA on LasHeR and RGBT234.

Variants	LasHeR			RGBT234	
	PR	NPR	SR	PR	SR
w/o TOMA	71.2	67.1	56.7	86.3	64.4
Euclidean Distance	71.7	67.5	57.1	87.0	64.8
w/o Barycenter Refinement	71.8	67.6	56.9	87.2	64.7
w/o Anchor Protection	71.6	67.8	57.0	86.9	65.0
w/o SVD Recovery	71.5	67.4	57.1	87.1	64.6
w/o Memory Threshold	71.4	67.6	56.8	86.8	64.9
Fixed Momentum	72.0	67.9	57.2	87.4	65.1
Full TOMA	72.2	68.2	57.4	87.6	65.3

4.4.2. Validation of the effectiveness of TOMA

To isolate the effect of test-time adaptation, all variants in Table 6 use the same trained spatial model with CD3 and TSFE enabled, while differing only in the test-time adaptation strategy. The non-adaptive variant, which disables TOMA updates, is adopted as the baseline. As shown in Table 6, TOMA achieves the best overall tracking accuracy compared with the non-adaptive baseline and representative Test-Time Adaptation methods, including AdaBN, Tent, EATA, and TCA.

While dynamic manifold optimization introduces computational overhead, TOMA maintains a favorable accuracy-efficiency balance. Evaluated on the LasHeR test set using a single thread, our framework runs at 50 FPS. Although slightly slower than the baseline and other TTA variants, it comfortably exceeds the real-time tracking threshold, ensuring practical applicability.

As shown in Table 7, replacing or removing any major component of TOMA leads to different degrees of performance degradation. The declines caused by Euclidean distance, removing barycenter refinement, and disabling anchor protection demonstrate the importance of distribution-aware state measurement, redundant-state compression, and stable reference preservation. The memory threshold and SVD recovery also contribute noticeably by preventing unreliable statistics from entering the manifold bank and recovering more stable adaptation directions. In contrast, fixed momentum results in a relatively smaller decrease, indicating that adaptive momentum mainly serves as an auxiliary optimization strategy. With all components enabled, Full TOMA achieves the best results of 72.2%/68.2%/57.4% on LasHeR and 87.6%/65.3% on RGBT234, confirming the complementary effects of its internal designs.

To further evaluate the distance metric used in TOMA, we compare the frame-wise center errors produced by Wasserstein and Euclidean distance. As shown in Fig. 12, the Wasserstein variant maintains lower and more stable localization errors on both sequences. On *baggirl*, the Euclidean variant suffers severe accumulated drift in the later stage, whereas the Wasserstein variant remains relatively stable; on *minibus*, the Wasserstein variant also achieves smaller errors over most frames. This comparison indicates that Wasserstein distance provides a more reliable measure of statistical distribution shifts and improves the robustness of test-time adaptation.

4.4.3. Hyperparameter sensitivity of CD3

To examine whether the empirical choices in CD3 cause excessive hyperparameter sensitivity, we vary d_0 , k , and $t_{\max}$ individually while keeping the remaining settings fixed. As shown in Table 8, the maximum variations are only 0.4%, 0.4%, and 0.3% in PR, NPR, and SR, respectively.

For d_0 , a smaller value triggers attenuation too early, whereas a larger one delays conflict suppression. Although $d_0 = 0.4$ slightly improves SR, $d_0 = 0.3$ achieves the best PR and NPR. For k , the default value of 8 performs best on all metrics, while smaller or larger values make the transition overly smooth or relatively steep. For $t_{\max}$, 3.0 achieves the best PR and SR and ties for the best NPR, indicating a better

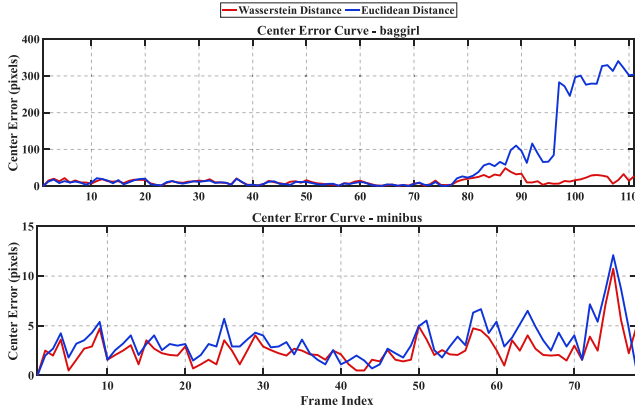


Fig. 12. Frame-wise center error comparison between Wasserstein and Euclidean distance in TOMA on two representative LasHeR sequences: *baggirl* and *minibus*.

Table 8

Hyperparameter sensitivity analysis of CD3 on LasHeR. Only one parameter is varied at a time, while the remaining parameters are fixed to their default values.

Hyperparameters		LasHeR		
Parameter	Value	PR	NPR	SR
d_0	0.2	71.8	67.9	57.2
d_0	0.3	72.2	68.2	57.4
d_0	0.4	72.0	68.0	57.5
k	4	72.0	68.1	57.2
k	8	72.2	68.2	57.4
k	12	72.0	67.8	57.3
$t_{\max}$	2.0	71.9	67.8	57.3
$t_{\max}$	3.0	72.2	68.2	57.4
$t_{\max}$	4.0	72.0	68.2	57.2

balance between conflict suppression and infrared information preservation.

Overall, the adopted setting, i.e., $d_0 = 0.3$, $k = 8.0$, and $t_{\max} = 3.0$, provides the best overall balance among the tested configurations, while the limited performance variations indicate that CD3 is not highly sensitive to these hyperparameters.

4.4.4. Ablation on CD3 and TSFE layers

We investigate the optimal network depth for the CD3 and TSFE modules, with the results summarized in Table 9. First, by fixing the CD3 module to 2 layers, we evaluate different configurations of TSFE. The results indicate that utilizing a single TSFE at layer 4 achieves the peak performance. Employing multiple TSFE layers or shifting the position of the single layer leads to performance degradation, likely due to feature redundancy or over-smoothing. Furthermore, when fixing the TSFE at layer 4, we observe that a 2-layer CD3 provides the optimal representational balance, outperforming both shallower and deeper variants. Therefore, we adopt 2 CD3 layers and a single TSFE at layer 4 as our default configuration.

4.5. Limitations

Despite its effectiveness in mitigating cross-modal interference and temporal drift, CoManTrack has a few notable limitations. First, the continuous distribution alignment and matrix decomposition required by the TOMA module introduce non-trivial computational overhead. This complexity may constrain real-time deployment on resource-limited edge platforms, such as UAVs. Moreover, the current implementation of TOMA relies on accessible BatchNorm statistics in the prediction head,

Table 9

Ablation study on the number of CD3 and TSFE layers on the LasHeR dataset.

CD3 Layers			TSFE Layers					LasHeR		
1	2	3	3	4	5	6	7	PR	NPR	SR
	✓		✓					71.8	67.8	57.1
	✓			✓				72.2	68.2	57.4
	✓				✓			71.4	67.3	56.7
	✓					✓		71.9	67.7	57.1
	✓						✓	71.5	67.4	56.7
	✓			✓	✓			70.7	66.9	56.4
	✓			✓			✓	70.6	66.9	56.2
✓				✓				70.8	66.9	56.2
		✓		✓				71.8	68.0	57.3

and its extension to architectures without such normalization statistics may require alternative distribution descriptors. Second, because the TSFE module captures absolute distributional boundaries to construct its contextual envelope, it is theoretically vulnerable to extreme hardware artifacts, such as uncorrected dead sensor pixels that artificially skew token extremums. Finally, the current CD3 module adopts a lightweight one-directional infrared-dampening design and is inserted only into an early ViT layer for efficient conflict control. Although effective for active thermal-distractor cases, this design may be less optimal under modality-reversal or joint-degradation scenarios, where the visible modality is severely degraded while infrared cues become more reliable. In such cases, the lack of bidirectional reliability estimation may lead to suboptimal conflict estimation and feature gating. Future work will explore lightweight manifold approximations and bidirectional reliability-aware conflict modeling to improve generality under diverse modality-degradation patterns.

5. Conclusion

In this paper, we propose CoManTrack, a spatio-temporally decoupled framework designed to tackle cascading cross-modal corruption and long-term drift in RGBT tracking. Spatially, we introduce Conflict-Driven Directed Dampening (CD3) to actively suppress high-energy thermal distractors, thereby protecting foundational visual structures. Building upon this purified feature space, the Tri-Statistic Feature Envelope (TSFE) captures the complete distributional boundaries of the multi-modal representations, providing a stable structural context that effectively prevents feature homogenization during deep aggregation. Temporally, we develop Test-Time Optimal Manifold Adaptation (TOMA) to handle target deformations. By dynamically evaluating the distributional novelty of incoming frames, TOMA filters redundant observations to maintain a diverse and well-conditioned target manifold, driving continuous and geometrically robust test-time statistical adaptation. Extensive evaluations across major benchmarks confirm that CoManTrack achieves competitive and balanced performance, demonstrating robust behavior under severe degradation. Moving forward, we plan to explore lightweight manifold approximations to further enhance its deployment efficiency on edge devices.

CRedit authorship contribution statement

Yujia Dong: Writing – original draft, Visualization, Methodology; **Haiyan Li:** Writing – review & editing, Supervision; **Yajie Liu:** Formal analysis; **Xun Lang:** Investigation; **Pengfei Yu:** Data curation; **Hao Zhou:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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